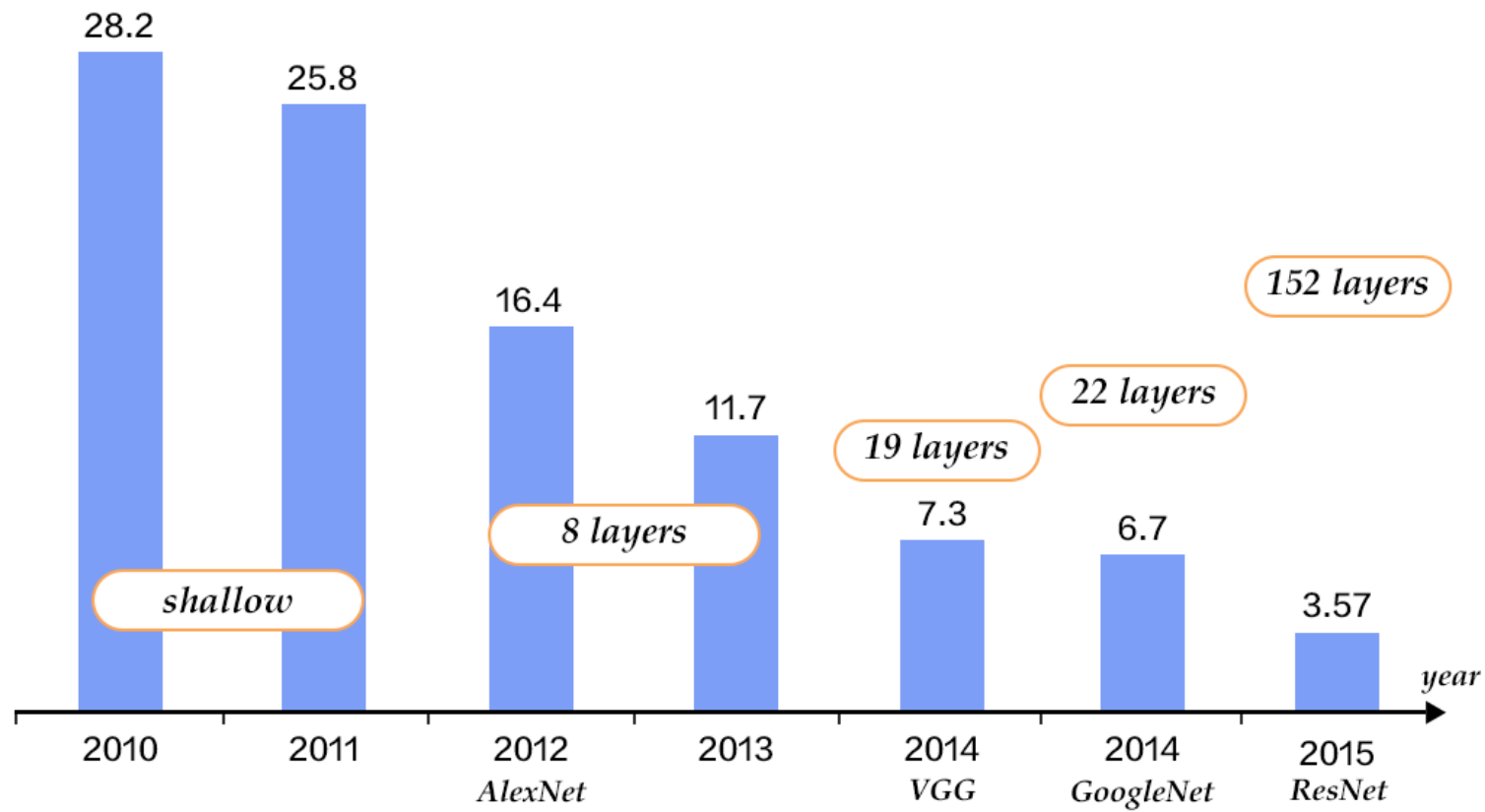


DEEP LEARNING

Архитектуры сверточных
сетей

AlexNet

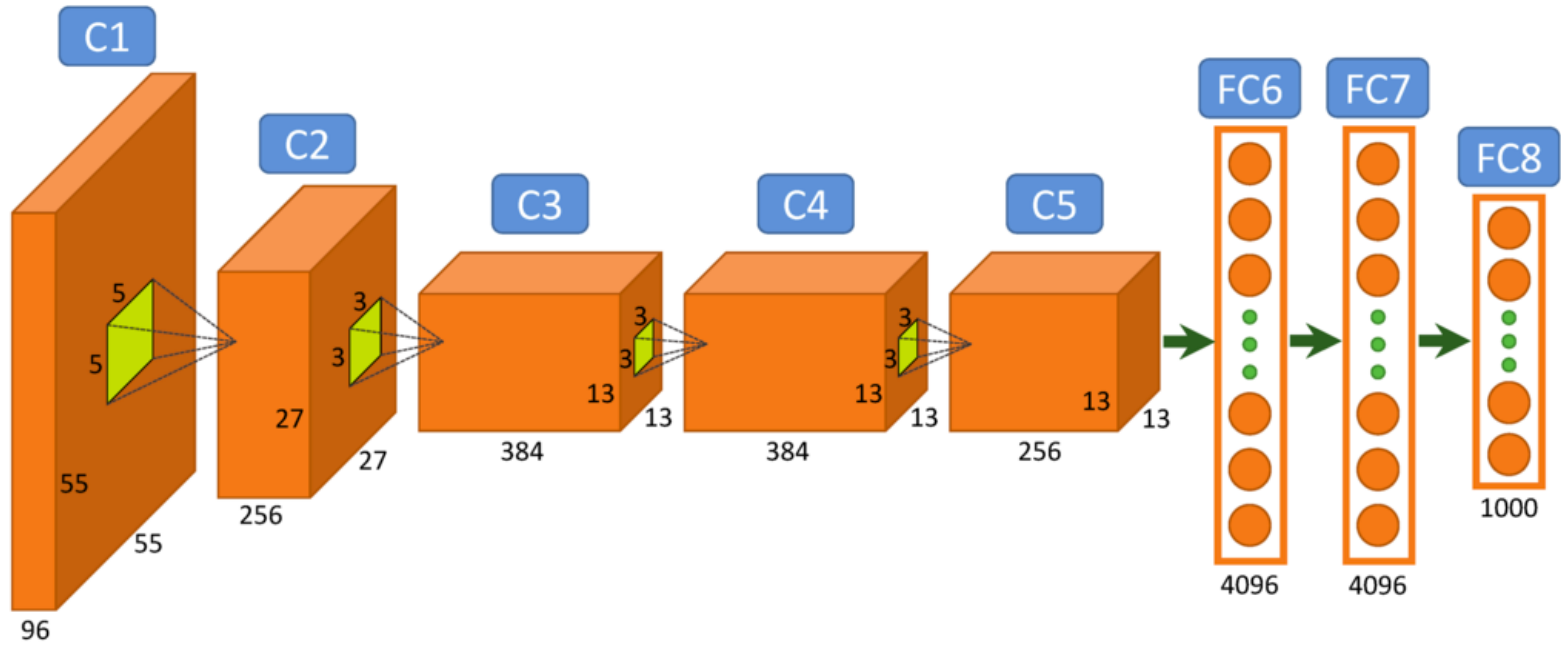
ILSVRC



Архитектура



AlexNet



GPU



Строительные блоки

Строительные блоки

Convolution

ReLU

Pooling

CrossChanel
Normalization

Строительные блоки

Convolution

FullyConnected

ReLU

Dropout

Pooling

CrossChanel
Normalization

Строительные блоки

Convolution

ReLU

Pooling

CrossChanel
Normalization

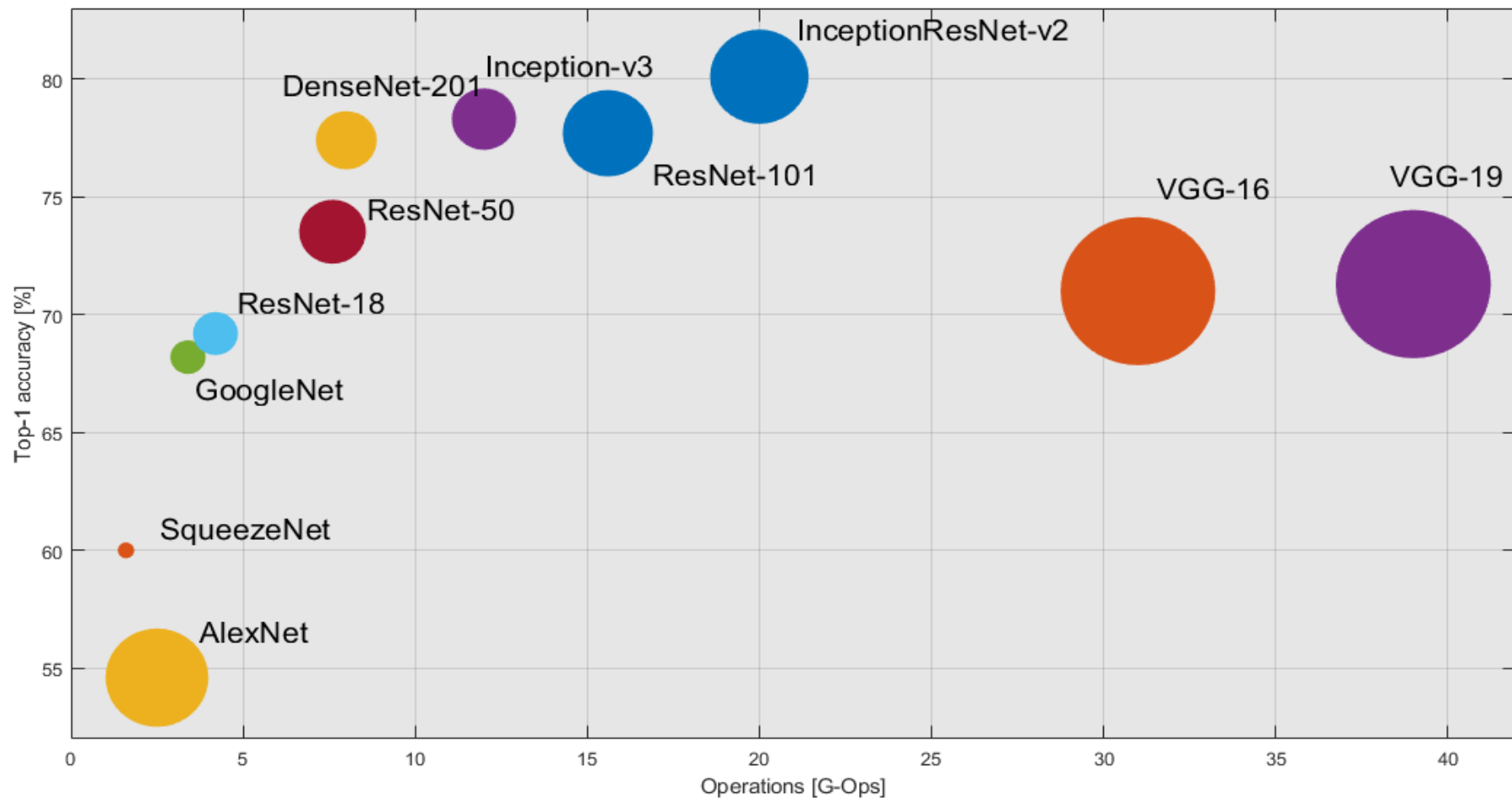
FullyConnected

Dropout

Input

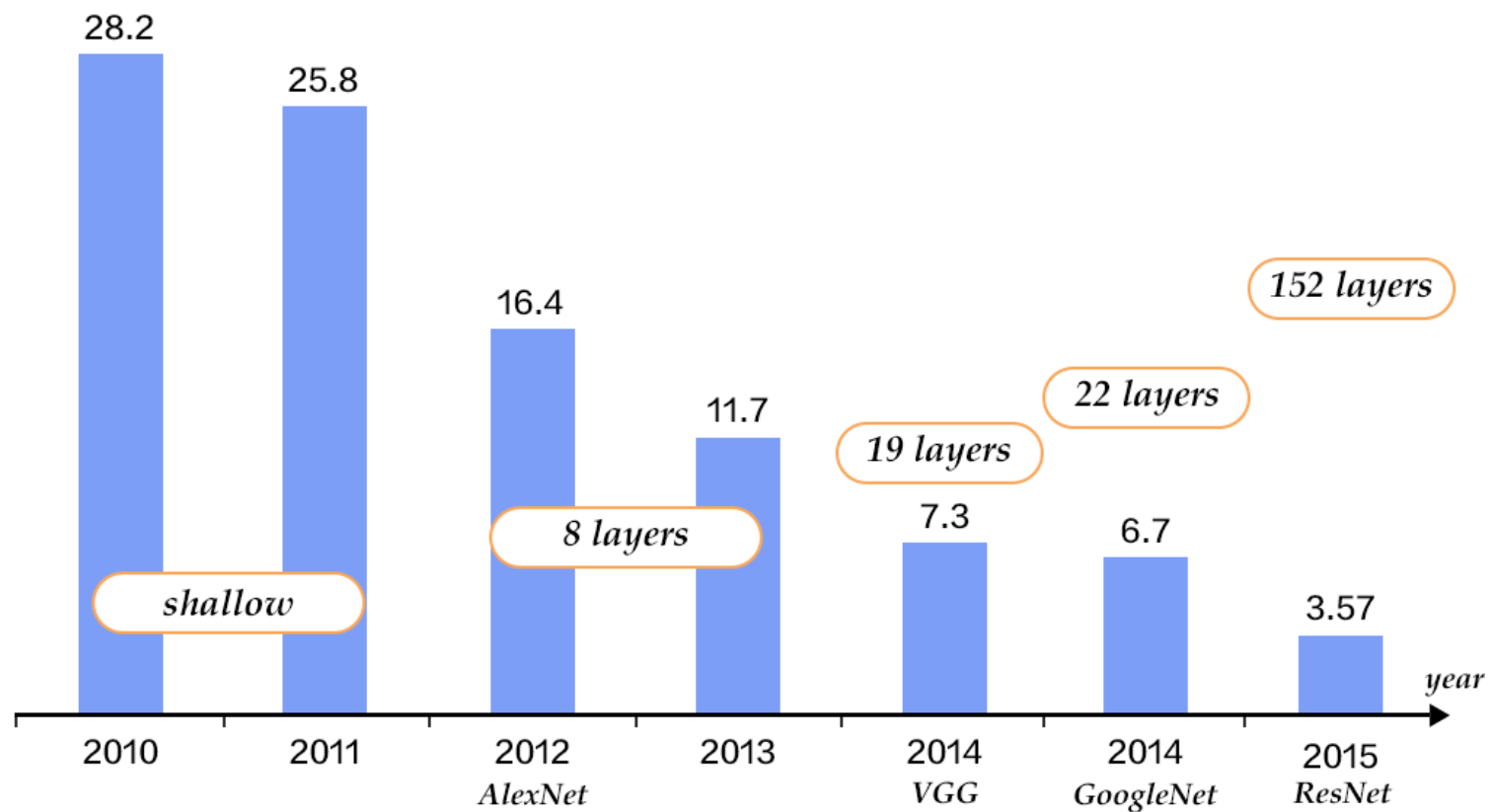
Softmax

Classification



VGG - 16 (19)

ILSVRC



Архитектура

VGG-16



Строительные блоки

Convolution

Input

FullyConnected

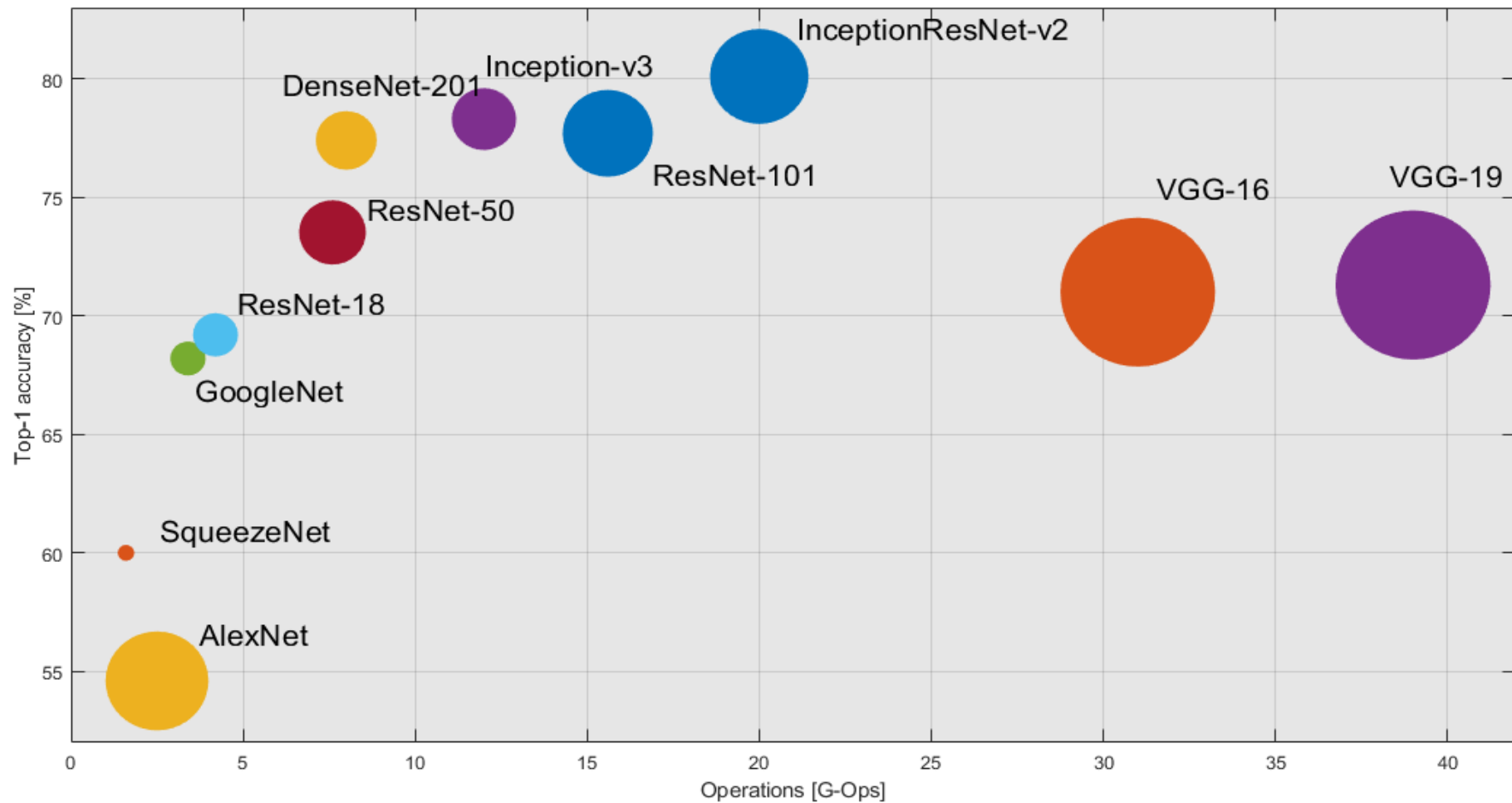
ReLU

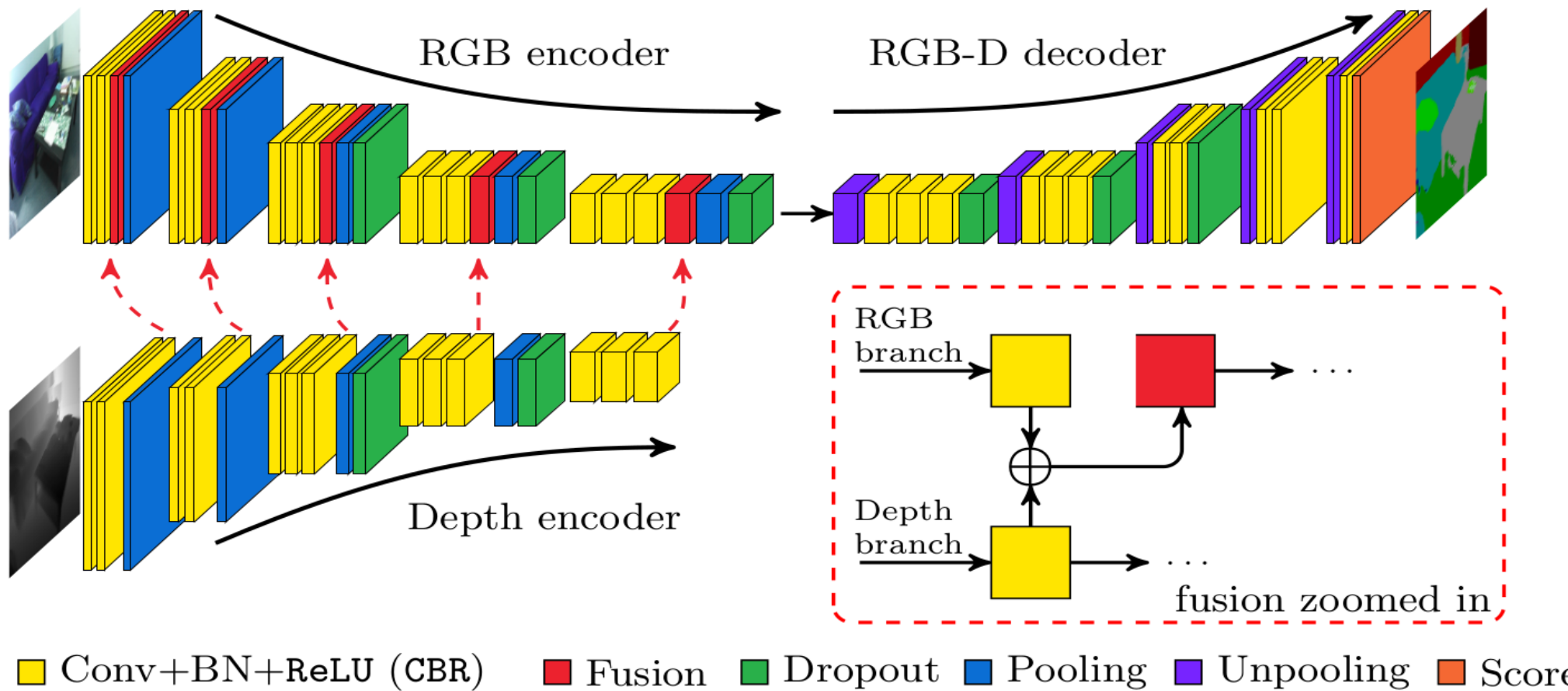
Softmax

Dropout

Pooling

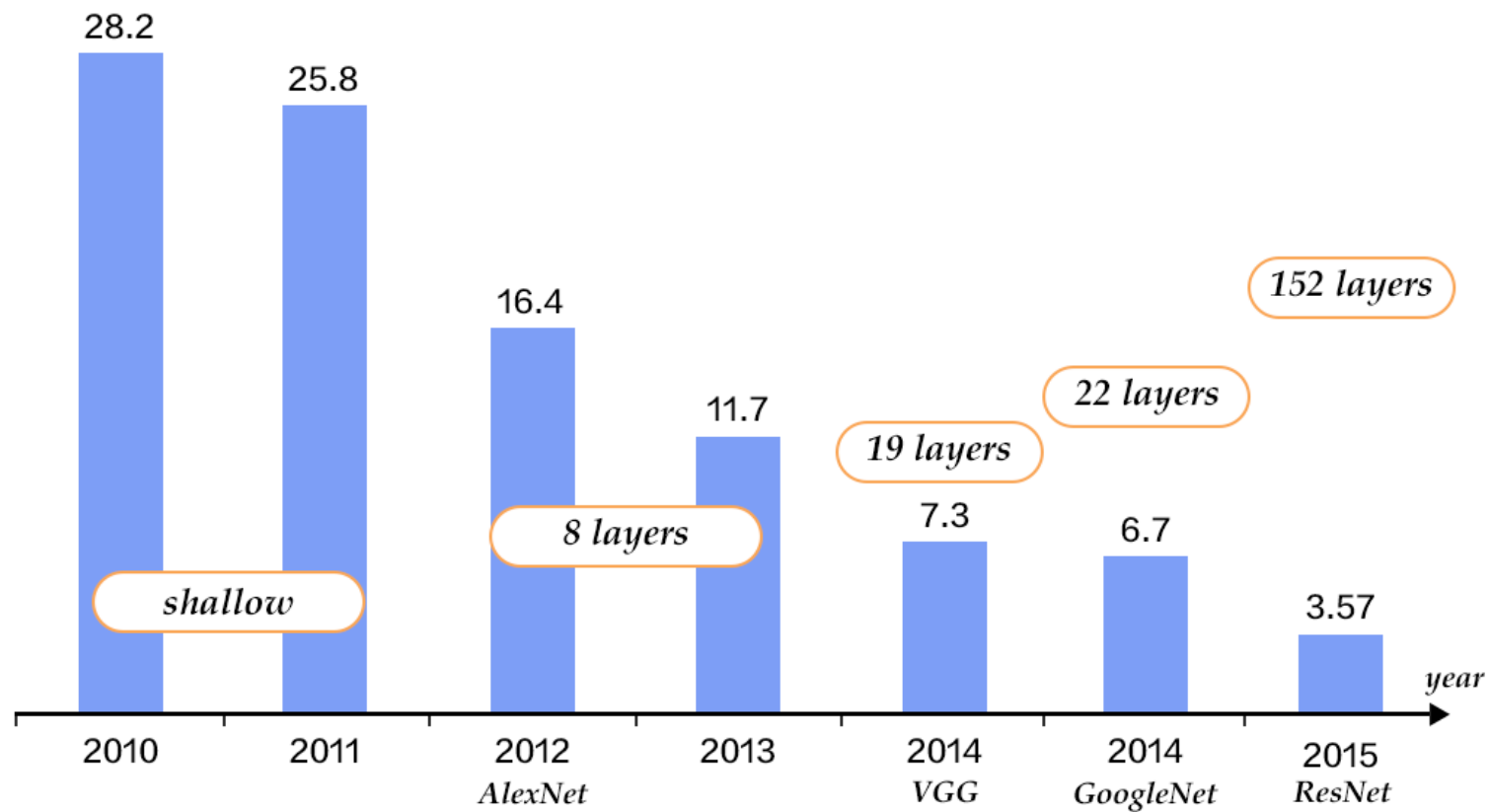
Classification



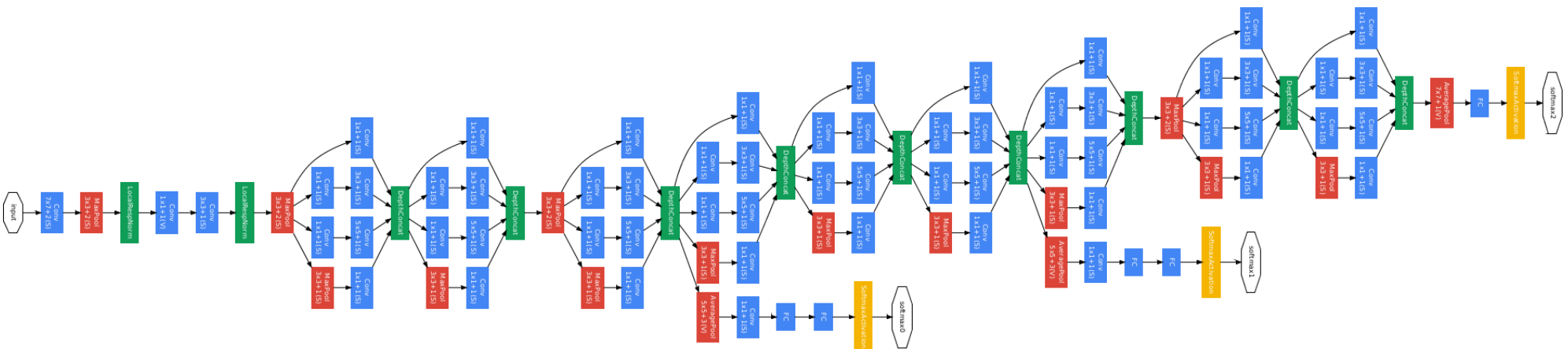


GoogleNet

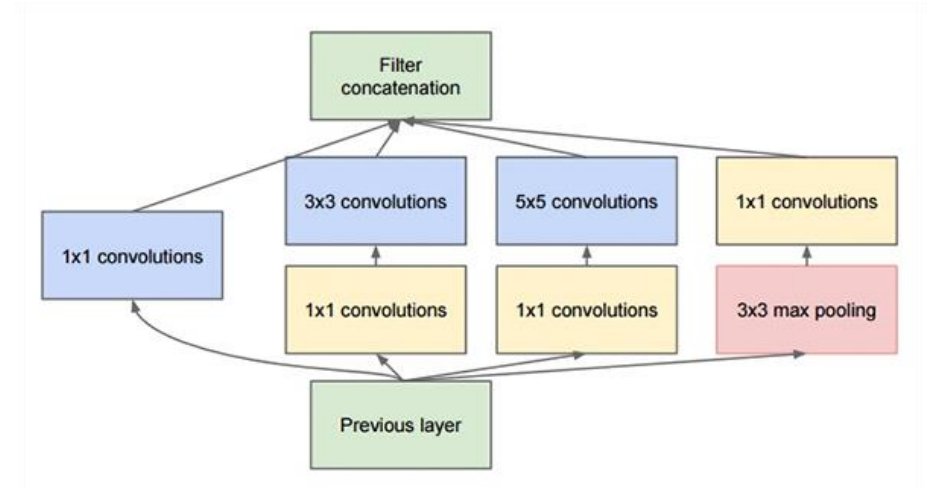
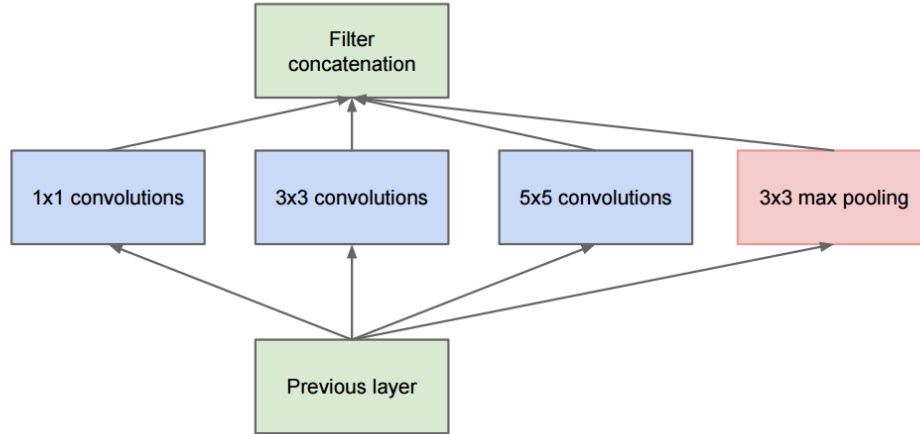
ILSVRC



Архитектура



GoogleNet (inception)



Строительные блоки

Строительные блоки

Convolution

ReLU

Pooling

CrossChanel
Normalization

Строительные блоки

Convolution

ReLU

Pooling

CrossChanel
Normalization

FullyConnected

Dropout

Строительные блоки

Convolution

ReLU

Pooling

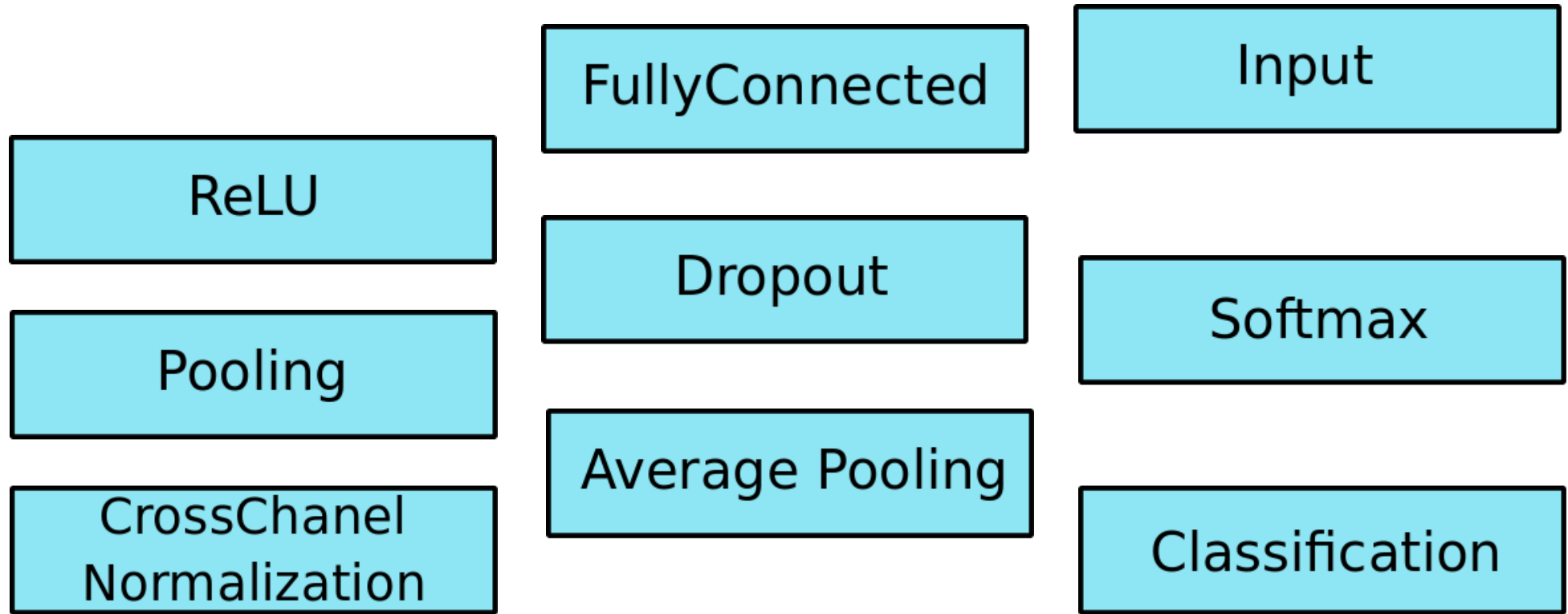
CrossChanel
Normalization

FullyConnected

Dropout

Average Pooling

Строительные блоки



Строительные блоки

Convolution

FullyConnected

Input

ReLU

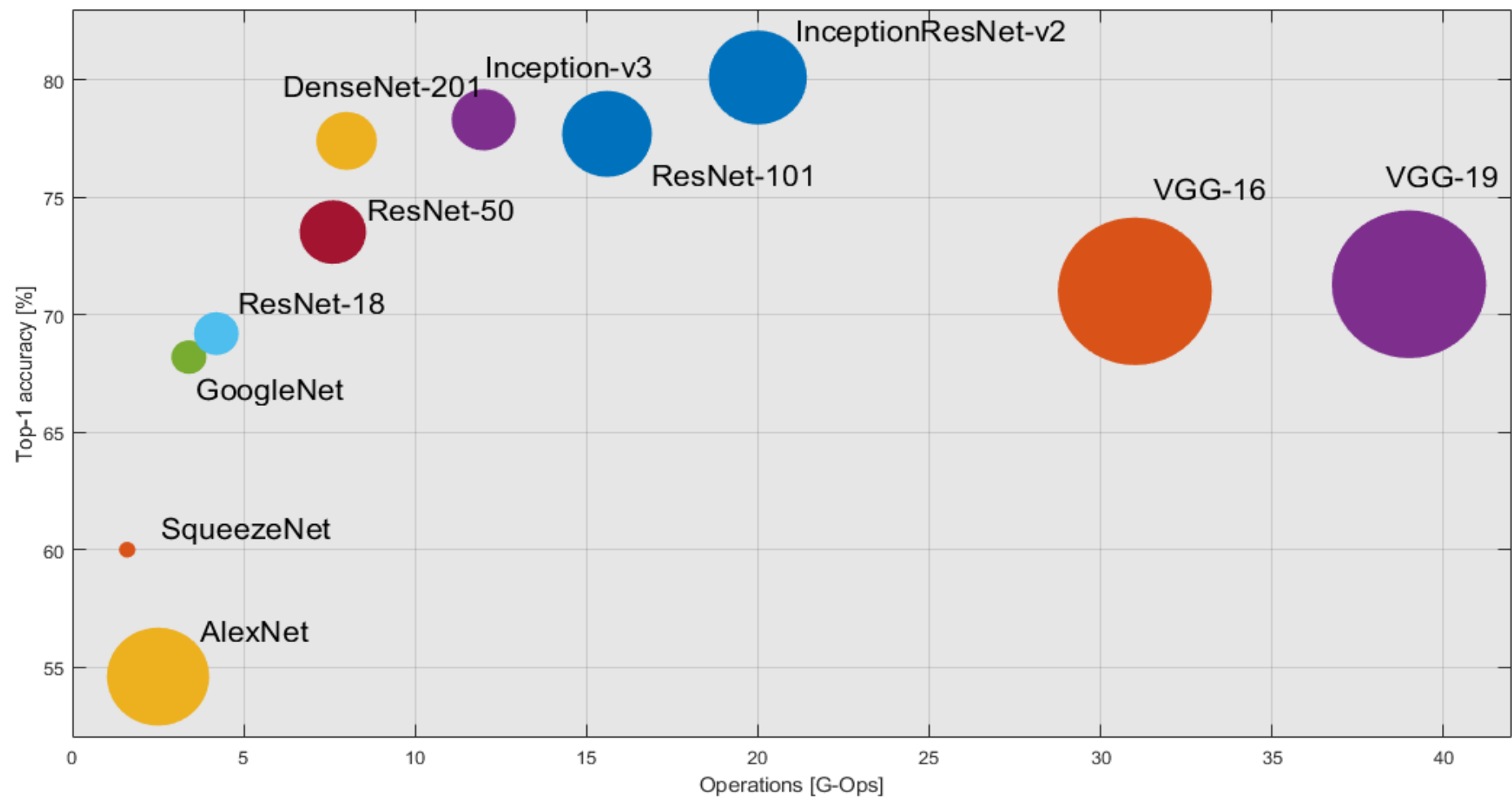
Softmax

CrossChanel
Normalization

Average Pooling

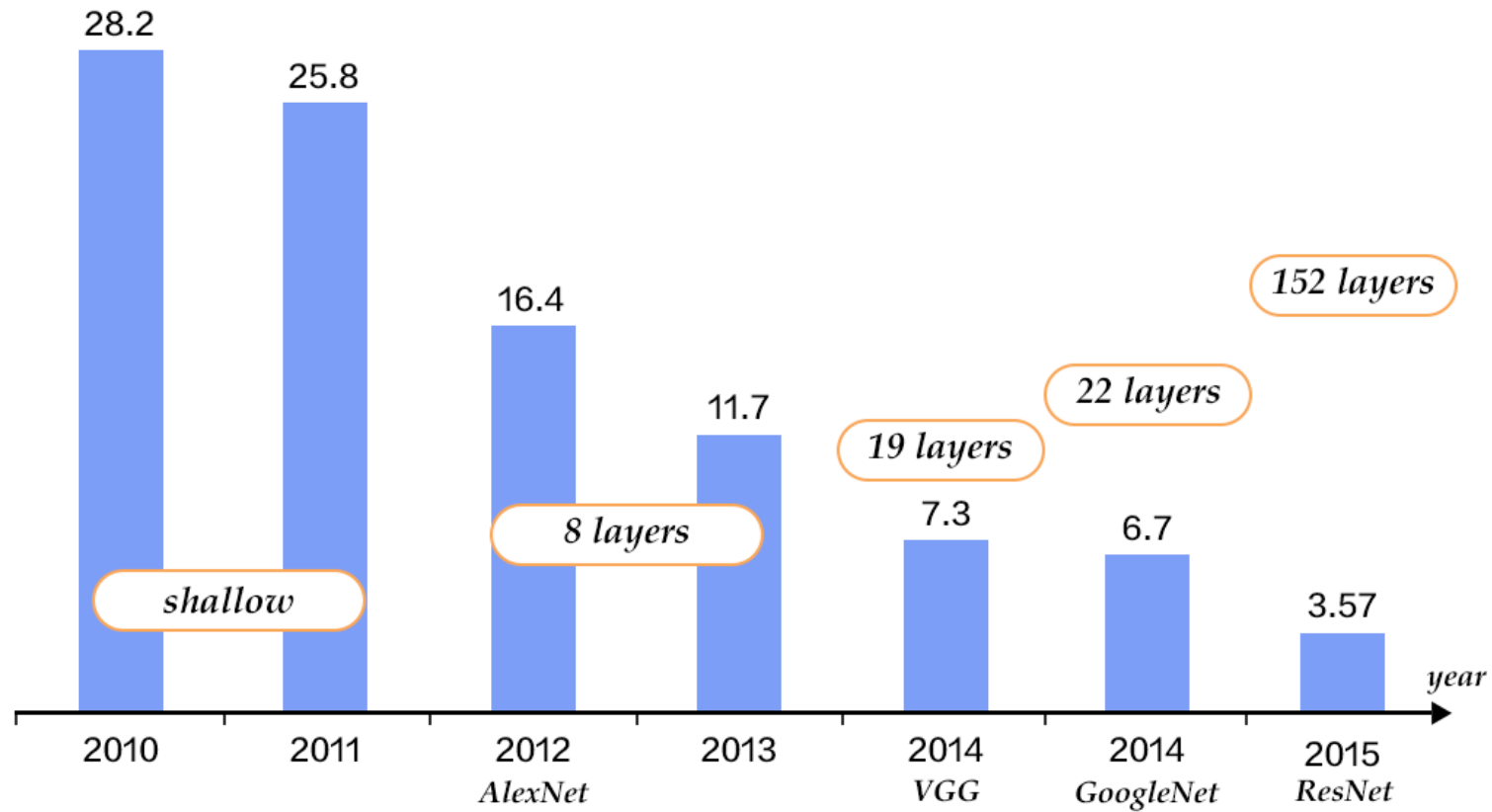
Classification

DepthConcatenation



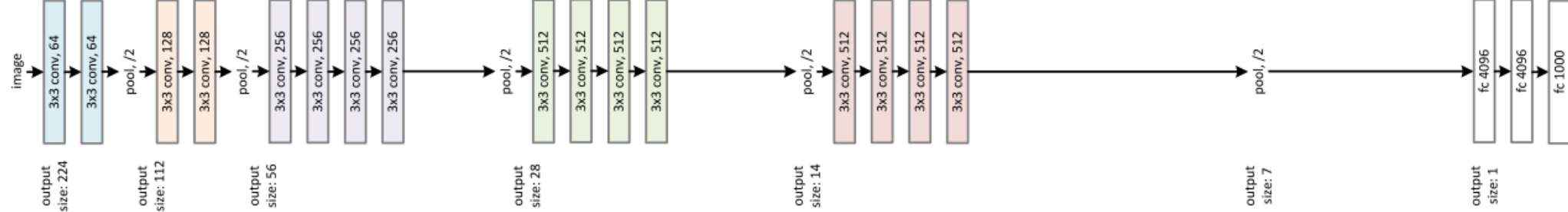
ResNet

ILSVRC

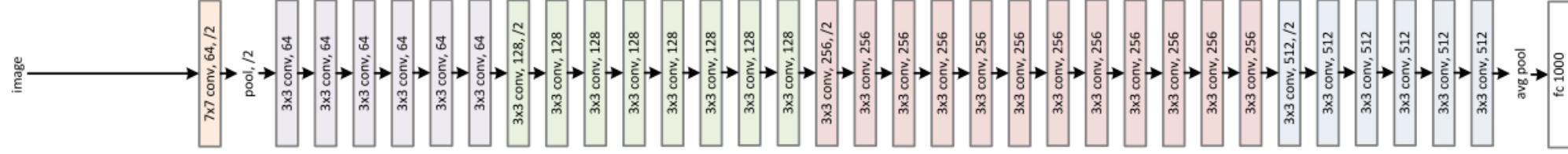


Архитектура

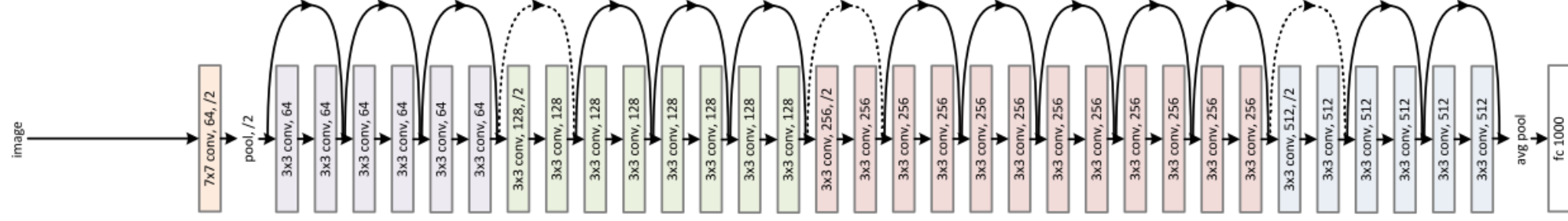
VGG-19

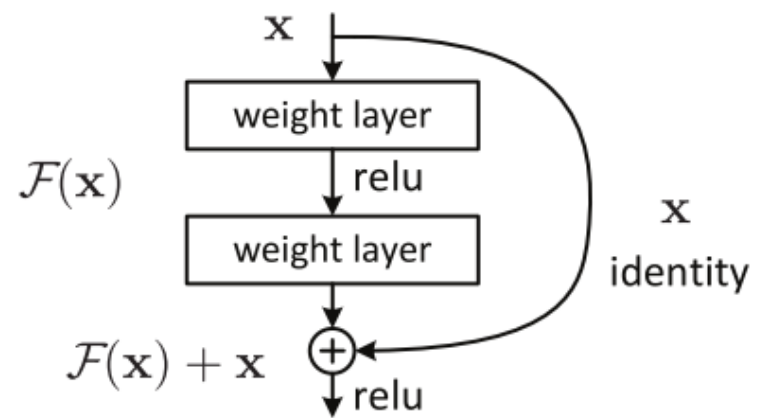


34-layer plain



34-layer residual





Строительные блоки

Строительные блоки

Convolution

ReLU

Pooling

Batch
Normalization

Строительные блоки

Convolution

ReLU

Pooling

Batch
Normalization

FullyConnected

Dropout

Строительные блоки

Convolution

ReLU

Pooling

Batch
Normalization

FullyConnected

Dropout

Average Pooling

Строительные блоки

Convolution

ReLU

Pooling

Batch
Normalization

FullyConnected

Dropout

Average Pooling

Input

Softmax

Classification

Строительные блоки

Convolution

ReLU

Pooling

Batch
Normalization

FullyConnected

Dropout

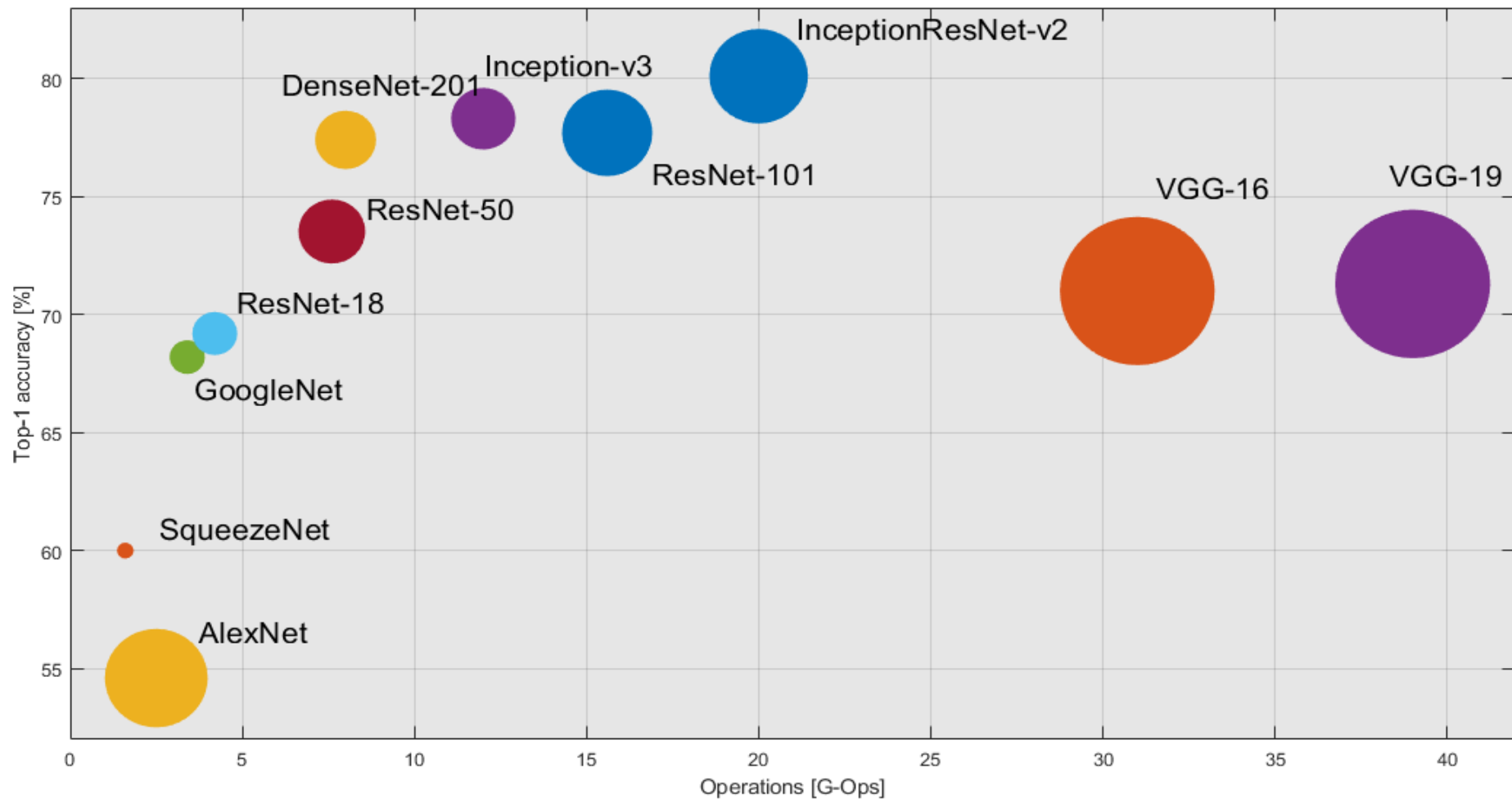
Average Pooling

DepthConcatenation

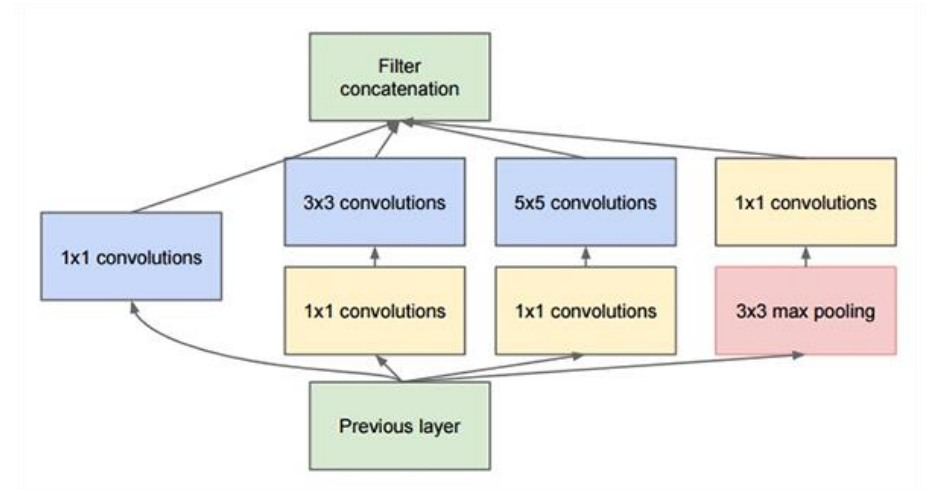
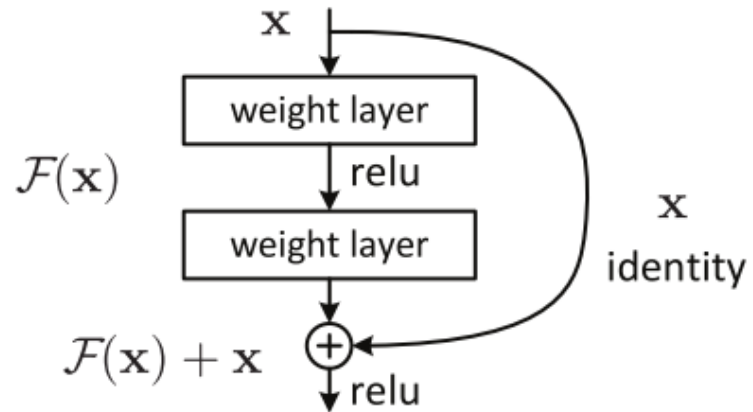
Input

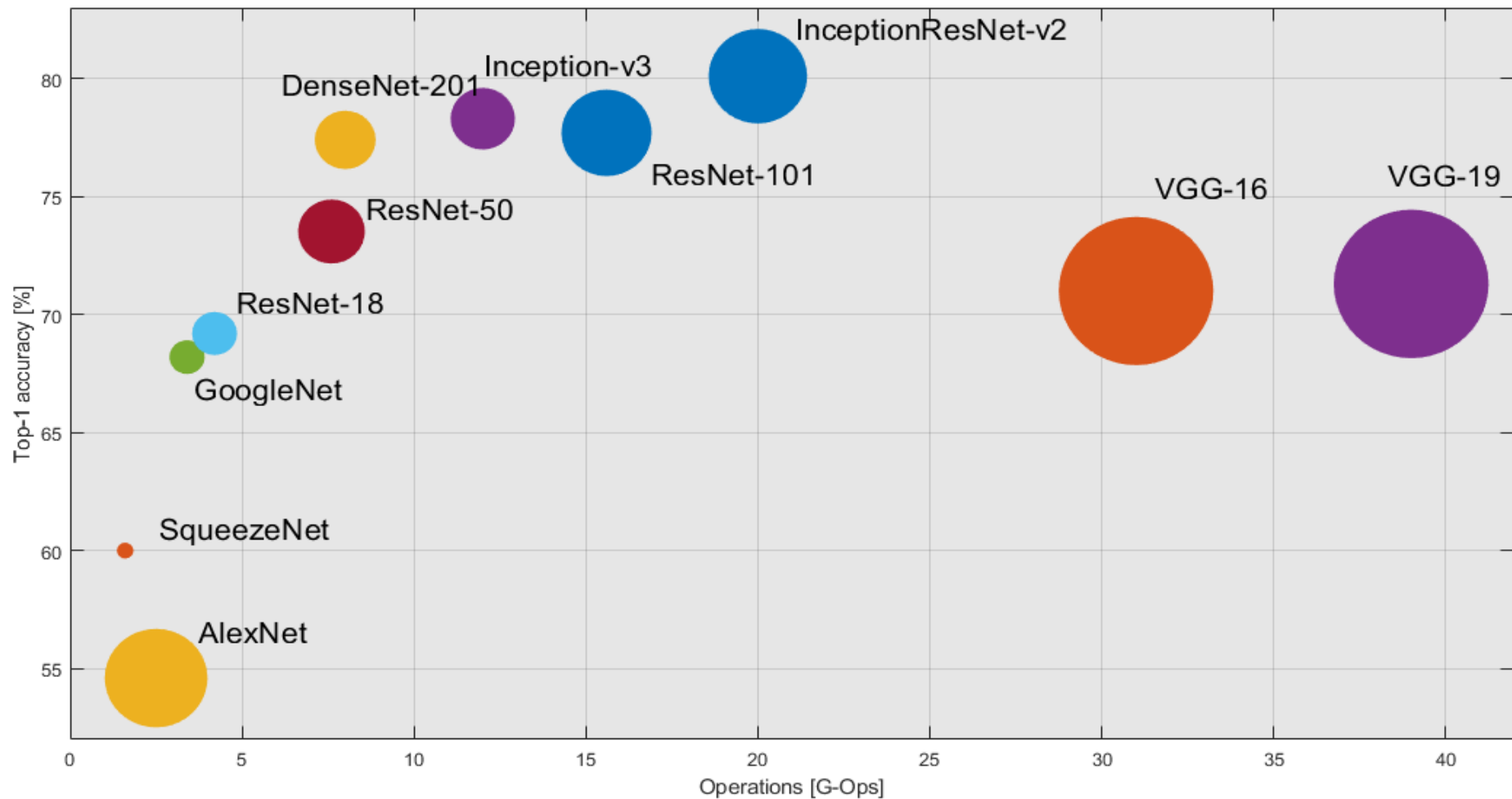
Softmax

Classification

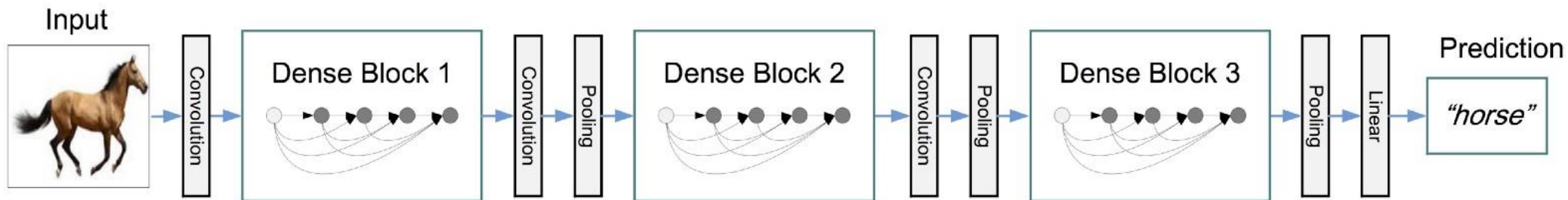
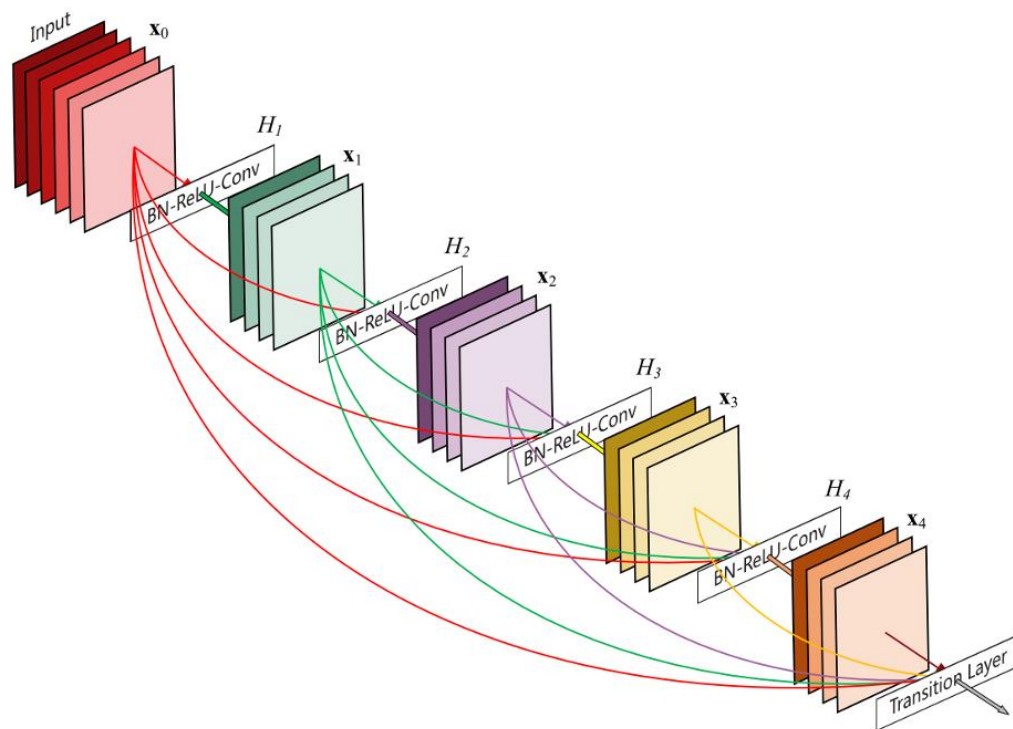


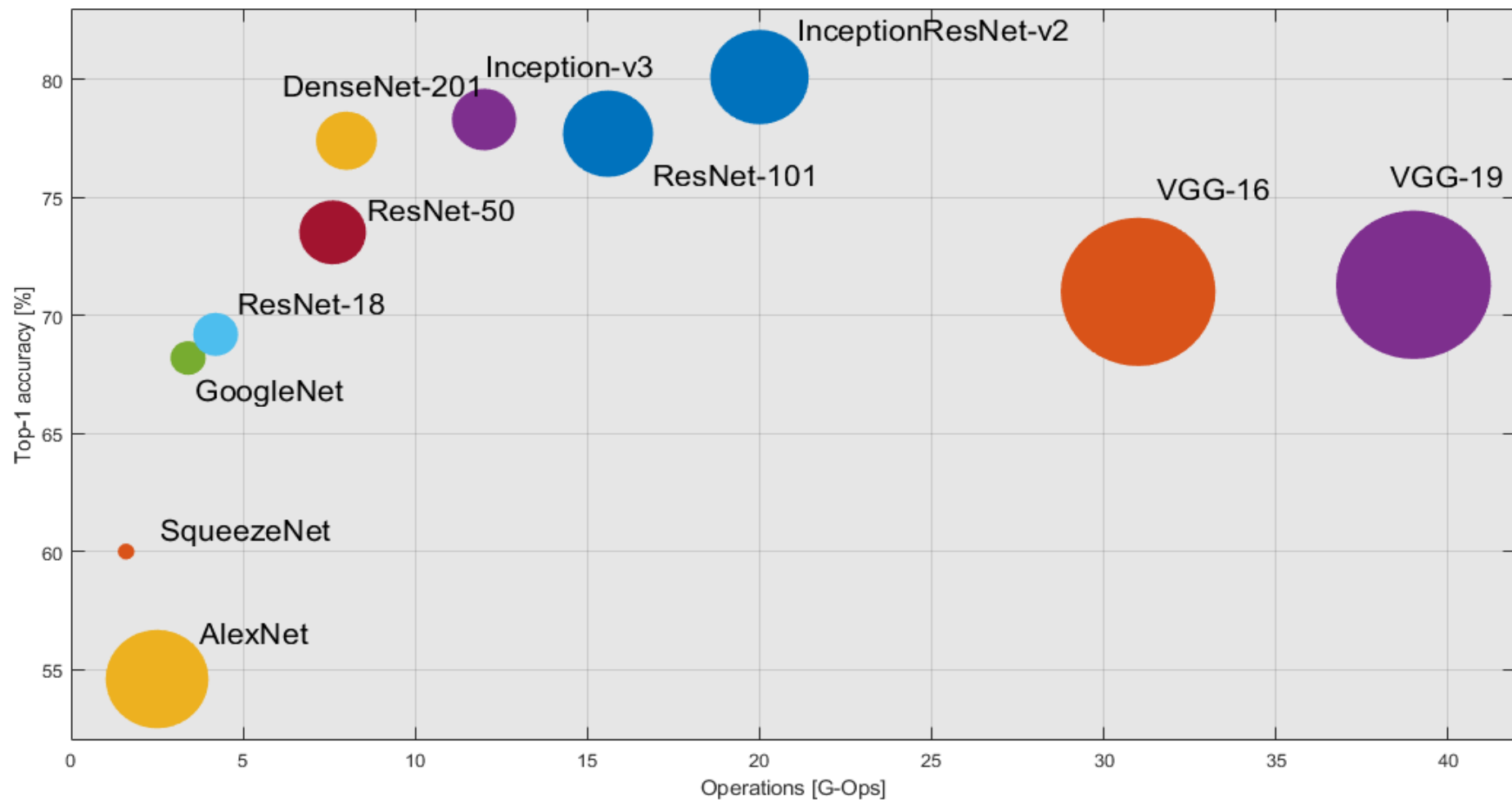
InceptionResNet





DenseNet





SqueezeNet

